KUKU FM • PS1 SYSTEM ARCHITECTURE & SOLUTION

Automated High-Retention Promo Discovery & Hook Optimization Engine

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Repository: ABHI-2802/kuku-promo-finder
Live Prototype: Cloudflare Verified

1. PROBLEM FRAMING: SOLVING COLD-START SOCIAL DROP-OFF

The Challenge: In micro-drama audio/video apps (TikTok, Reels, YouTube Shorts, Kuku FM feed), user retention follows a power-law drop: over 70% of viewers scroll past within the first 3.0 seconds unless an immediate, high-stakes hook commands attention. Furthermore, short-form promos (20–60s) cannot function as conventional summaries; they must operate as *high-curiosity tension engines* that build dramatic conflict and terminate abruptly on a cliffhanger, compelling the viewer to open the main app to discover the resolution.

Core Editorial Heuristics Engineered:

- **3-Second Hook Velocity:** Promos must open on high-voltage dialogue (threats, accusations, wedding interruptions) or visceral visual cues (bloodied garments, weapon reveals, frozen shock), avoiding slow atmospheric establishing shots.
- **Zero Spoilers / Tension Asymmetry:** Out-points must isolate the *unanswered question* without leaking the subsequent revelation.
- **Centisecond Boundary Precision:** Preserving natural speech cadence (`[a, b]` centisecond timecodes) ensures no mid-word audio clipping.

2. DUAL-ENGINE ARCHITECTURE & PRODUCTION SCALING

The system employs a resilient two-tier architecture tailored for high-throughput studio production and human-in-the-loop review:

Component	Offline Deterministic Rules Engine	Gemini AI Cognitive Mode (Dual-Pass)
Primary Purpose	Ultra-fast, zero-cost deterministic baseline utilizing regex pattern matching, acoustic pacing filters, and centisecond boundary validators.	Pass 1: Episode subtext analysis (emotional stakes, hook velocity, question extraction). Pass 2: Show-level cross-episode global ranking.
Hook Evaluation	Scans opening 3.0s (`a <= t <= a+300 cs`) for conflict triggers (e.g., 'stop this wedding', 'lies', 'blood', 'betrayal').	Evaluates cognitive curiosity gap, visceral surprise, and emotional stakes for an uninitiated, cold viewer.
Cliffhanger Logic	Identifies terminating spoken questions or visual consequence beats; cross-checks subsequent dialogue to verify mystery preservation.	Ranks unresolved dramatic queries, evaluates whether the audience will immediately crave episode unlocking.
Cost & Throughput	**$0.00 / run**. Processes 50 episodes in ~0.65 seconds (75+ eps/sec). Zero external API dependencies, 100% offline uptime.	**~$0.003 / episode** via Gemini 2.5 Flash (~$7.50 annual cost for 50 shows × 50 episodes = 2,500 scripts). Parallelized async queues.

3. BOUNDARY VALIDATION & OBSERVED FAILURE CASE STUDY

Compare Endings Mechanism: An automated safety inspector examines every candidate cut-point by retrieving the *earliest spoken line starting at or after the cut time* (`start >= cut_end_cs`). This displays both sides of the edit to editorial supervisors.

Observed Failure (Episode 12 Timing Desynchronization):

- *The Anomaly:* In Episode 12, a visual action cue (Cue 44: 'Camera sweeps across the gallery') terminates at `01:09.50`, while the preceding dialogue line (Cue 43) ended at `01:05.40`. A naive sequential index rule (`cue_id > end_cue_id`) incorrectly grabbed Cue 45 (`01:05.40`), which actually occurred *before* the visual cut.
- *The Failure Resolution:* We re-architected `earliest_spoken_after()` to strictly filter on absolute timecodes (`c['a'] >= cut_end_cs`). The validator now correctly surfaces Cue 48 (ALEXANDER: *'Juliana, no. Do not let them hurt you. I can handle the law.'*) starting at `01:09.80` (+0.30s after cut). This prevents premature cuts and misaligned subtitles.
- *Spoiler Leakage Catch (Episode 1):* Episode 1 terminates at `00:39.00` on the wedding interruption. The next line at `00:39.10` reveals the wife's identity (*'Because he is marrying me.'*). The validator verified that stopping at `00:39.00` preserves 100% cliffhanger potency without leaking the mystery.

4. FULL 50-EPISODE RUN RESULTS & VERIFICATION

The complete 50-episode private script collection (`private_scripts/`, SHA-256 fingerprint: `36920cb48b...`) was executed through the pipeline. **Run Metrics:** Exactly **50 episodes analyzed**, **50 promo cuts generated**, **0 rejected/passed** (100% yield). Durations strictly adhere to the 20.0s – 60.0s broadcast standard (Mean: 42.6s, Min: 21.0s, Max: 58.0s). All 50 cuts were validated against dialogue boundary violations.

Top Ranked Promo Cuts (Show-Level Global Standouts):

Rank	Ep / Timecode	Dur	0:00–0:03 Retention Hook	Unanswered Dramatic Question / Cliffhanger	Strategic Rationale
#1	**E36** 01:43.50–02:21.50	38.0s	*'Close-up on Alexander, his expression freezing in complete shock.'*	**'When?'** (Spoken query at 02:21.50)	Instant visual shockwave hook; isolates high-stakes temporal contradiction before explanation is revealed.
#2	**E3** 00:08.50–00:40.40	31.9s	*'Medium close-up on Juliana looking across at her sister with wounded pride.'*	**'And what happens to me?'** (Juliana at 00:40.40)	Direct sibling betrayal conflict; terminates on existential personal vulnerability.
#3	**E10** 00:06.80–00:53.30	46.5s	*'Alexander struggling to breathe, clutching bloodied coat over chest.'*	**'Does it hurt?'** (Spoken question at 00:53.30)	Visceral physical stakes inside first 3s; cuts out before medical or romantic confession.
#4	**E13** 01:26.50–01:58.88	32.4s	*'Juliana!'* (Shouted urgent call)	**[Visual Cliffhanger]** Macro close-up on Juliana's bloodied fist gripping emerald ring on frozen ground.	High acoustic hook; ends on symbolic mystery object without revealing attacker or fatality.
#5	**E26** 00:55.80–01:45.00	49.2s	*'Medium close-up of Duchess, voice trembling with tragic certainty.'*	**'Your Grace? What is wrong? You look as though you've seen a ghost.'**	Opens on royal matriarch distress; ends on haunting question demanding narrative payoff.

5. VERIFIED DELIVERABLES & SUBMISSION INTEGRITY

Verification Suite (`test_app.py`): Automated test suite passed all 5 regression tests:

1. *Parser Integrity:* Accurately handles multi-speaker dialogues, direction cues, and centisecond brackets.
2. *Boundary Timing:* Re-verified Episode 12 (`01:09.50` ➔ `01:09.80`) and Episode 1 (`00:39.00` ➔ `00:39.10`).
3. *Dual-Engine Evaluation:* Verified offline rules fallback and AI prompt schema compliance.
4. *50-Episode Pack Run:* 50/50 promos ranked and synchronized to export files.
5. *Fresh Upload Path:* In-memory ZIP parsing mirrors direct script folder execution with zero drift.

Deliverable File	Format & Path	Contents & Description
Ranked Promos CSV	`submission_outputs/ranked_promos.csv`	All 50 ranked promo segments with Hook, Cliffhanger, Timecodes, and Story-Specific 'Why It Works'.
Episode Coverage CSV	`submission_outputs/episode_coverage.csv`	50 rows validating 100% episode coverage, promo inclusion status, cue counts, and duration checks.
Full Run JSON	`submission_outputs/promo_run.json`	Complete structured run payload including show metadata, raw segment cues, and ranking rationales.
Live Prototype	Cloudflare Tunnel / Streamlit	Interactive UI featuring Episode Analytics, 9:16 Video Feed Simulator, and Compare Endings Inspector.

Security & Governance: `private_scripts/` and proprietary API keys are strictly excluded from source control via `.gitignore`. The system runs self-contained, reproducible, and ready for immediate studio deployment.